WOLDIA UNIVERSITY

INSTITUTE OF TECHNOLOGY

School of Computing

Department of Software Engineering

Course Title: Web Service

Course code: Seng5127

Individual Assignment: ShapeTool SOAP Web Service

GitHub Repository: https://github.com/Belete21-5/web-service-ass-1

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Project Overview

The **ShapeTool Web Service** is a SOAP-based provider developed using Spring Boot. It provides automated surface area calculations for five geometric shapes: Circle, Square, Rectangle, Parallelogram, and Triangle. The project follows a **Contract-First** design pattern, ensuring that the service interface is strictly defined before implementation.

Almost I described at the reademe file.

The core of the service is defined in shapes.xsd. To ensure data integrity, a custom restriction was implemented to prevent negative dimensions.

- **Custom Type:** nonNegativeDouble
- **Constraint:** xs:minInclusive value="0"
- **Target Namespace:** http://woldia.edu/shapetool

This design ensures that any request containing invalid data is rejected at the XML validation stage, protecting the server-side logic from processing mathematically impossible values.

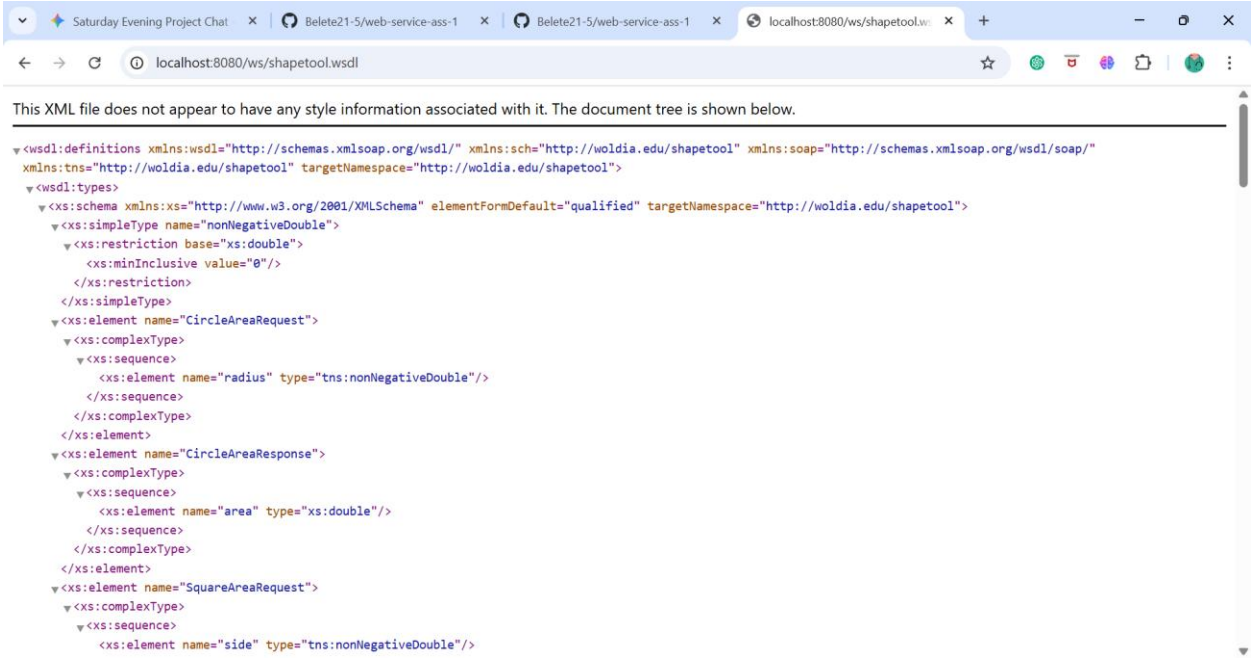
The service is built on the following architectural components:

- **WebServiceConfig.java**: Implements WsConfigurer to register the PayloadValidatingInterceptor. This component links the XSD schema to the request pipeline to enforce validation.
- **ShapeToolEndpoint.java**: Acts as the Service Provider. It uses JAXB-generated classes to handle incoming SOAP requests and applies the following formulas:
    - **Circle**: \pi \times r^2
    - **Rectangle**: length \times width
    - **Triangle**: 0.5 \times base \times height
    - Square: Area = side^2
    - Parallelogram: Area = base \times height

Testing and Validation

WSDL Generation

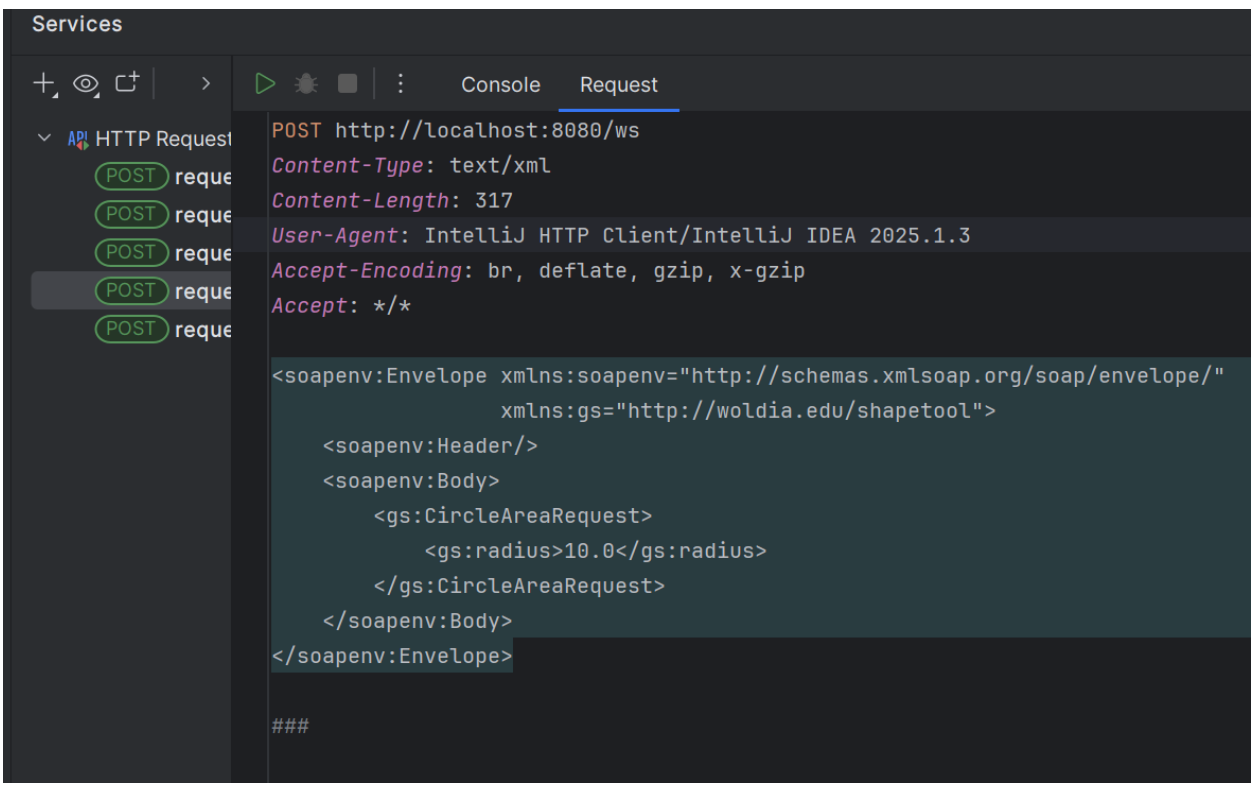
The service successfully generates a WSDL (Web Service Description Language) file, which acts as the formal contract for clients.

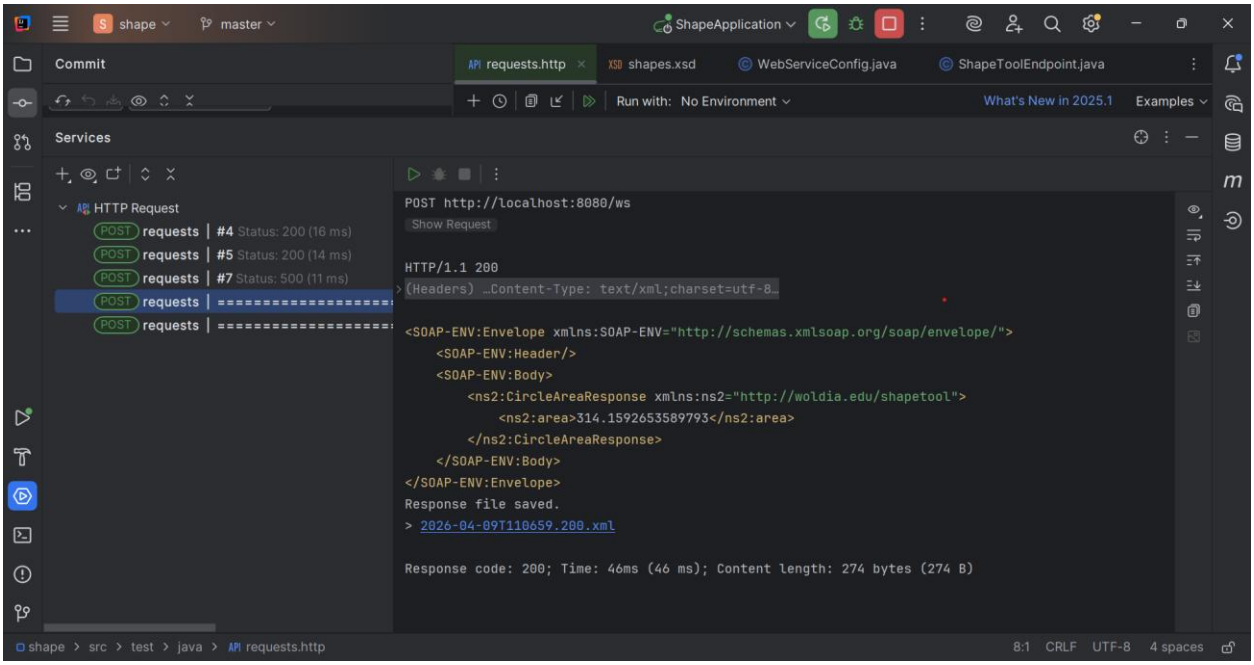


Functional Testing (Positive Cases)

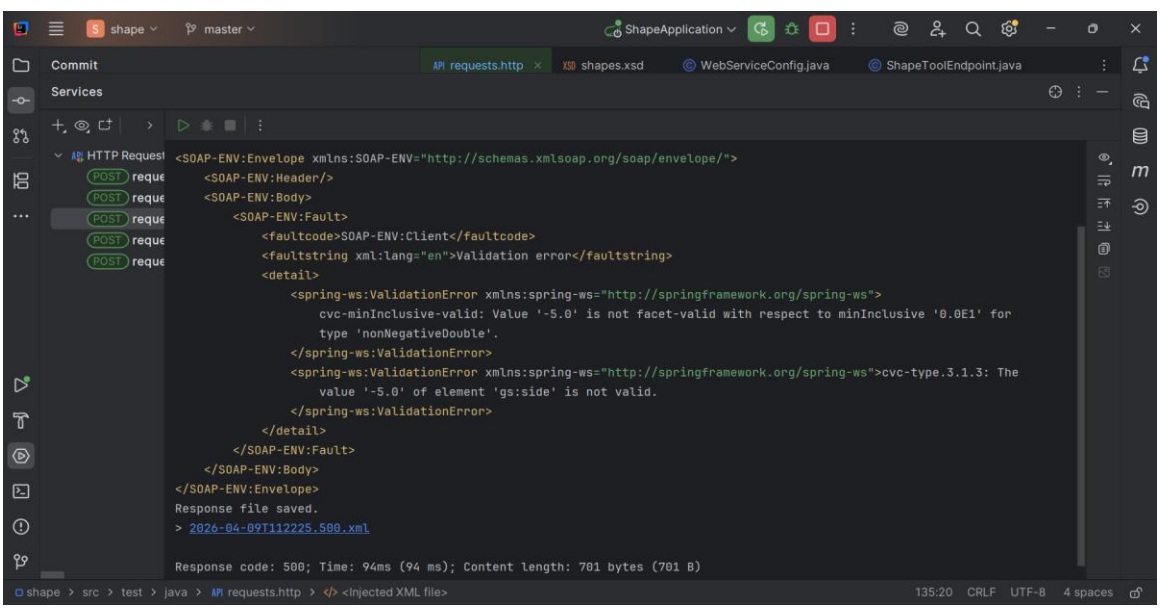
Using the IntelliJ HTTP Client, all shape calculations were verified.

- **Example (Circle):** Input Radius 10.0 → Output Area 314.159...





Input Validation (Negative Case)

To test the robust error handling, a negative value was sent (e.g., <side>-5.0</side>). The PayloadValidatingInterceptor correctly identified the schema violation.

- **Expected Result:** SOAP Fault (Validation Error)
- **Actual Result:** The service returned a Client Fault explaining that the value -5.0 is not valid per the XSD restrictions.